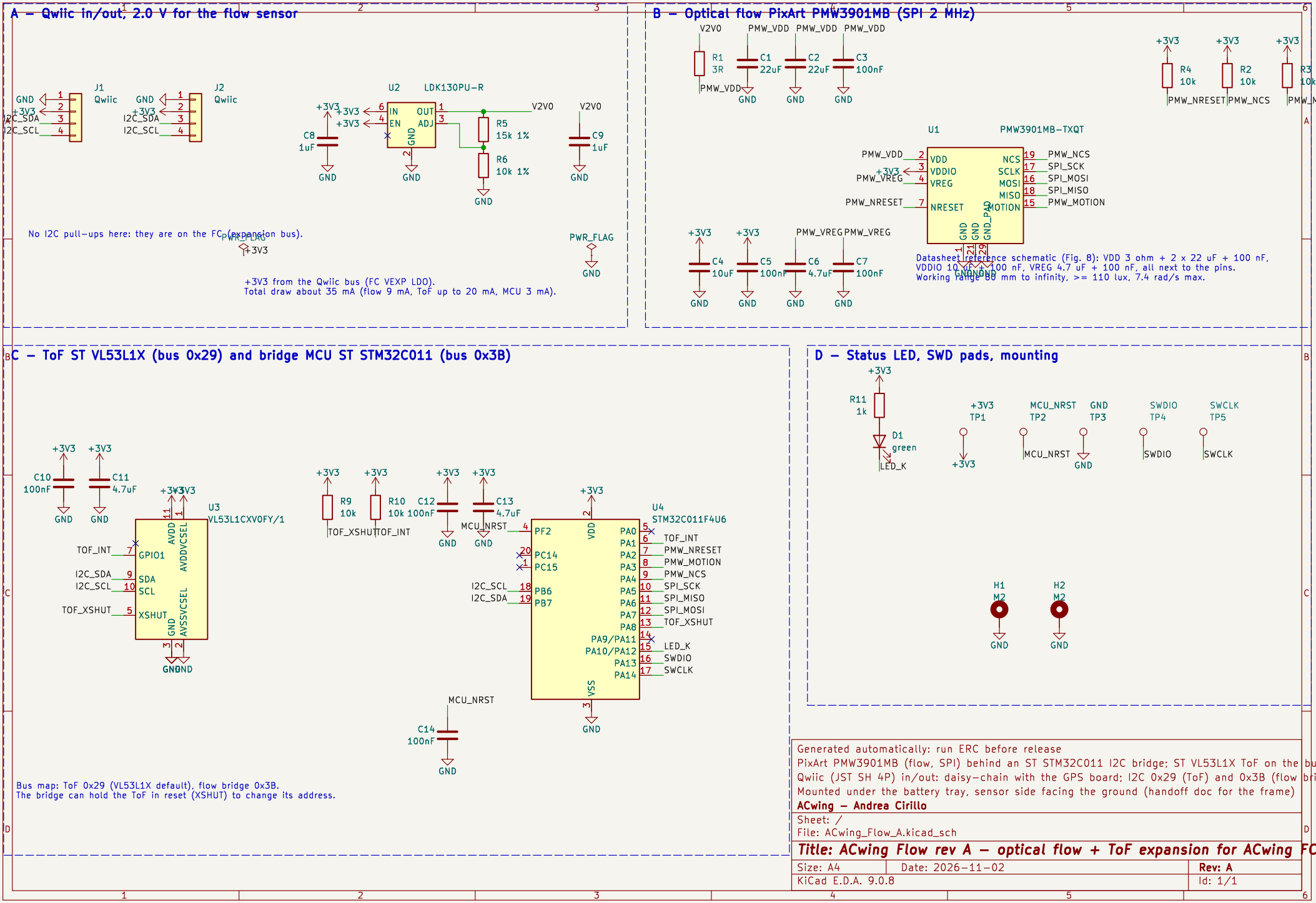

A — Qwiic in/out, 2.0 V for the flow sensor
J1
Qwiic
J2
Qwiic
GND
+3V3
I2C_SDA
I2C_SCL
U2
LDK130PU-R
IN
OUT
EN
ADJ
GND
V2V0
C8
1uF
C9
1uF
R5
15k 1%
R6
10k 1%
No I2C pull-ups here: they are on the FC (expansion bus).
PWR_FLAG
+3V3 from the Qwiic bus (FC VEXP LDO).
Total draw about 35 mA (flow 9 mA, ToF up to 20 mA, MCU 3 mA).
B — Optical flow PixArt PMW3901MB (SPI 2 MHz)
R1
3R
C1
22uF
C2
22uF
C3
100nF
PMW_VDD
R4
10k
R2
10k
R3
10k
PMW_NRESET
PMW_NCS
PMW_MOTION
U1
PMW3901MB-TXQT
VDD
VDDIO
VREG
NRESET
NCS
SCLK
MOSI
MISO
MOTION
GND_PAD
PMW_VREG
SPI_SCK
SPI_MOSI
SPI_MISO
C4
10uF
C5
100nF
C6
4.7uF
C7
100nF
Datasheet reference schematic (Fig. 8): VDD 3 ohm + 2 x 22 uF + 100 nF,
VDDIO 10 uF + 100 nF, VREG 4.7 uF + 100 nF, all next to the pins.
Working range 80 mm to infinity, >= 110 lux, 7.4 rad/s max.
C — ToF ST VL53L1X (bus 0x29) and bridge MCU ST STM32C011 (bus 0x3B)
C10
100nF
C11
4.7uF
U3
VL53L1CXV0FY/1
AVDD
AVDDVCSEL
AVSSVCSEL
GPIO1
SDA
SCL
XSHUT
TOF_INT
TOF_XSHUT
R9
10k
R10
10k
C12
100nF
C13
4.7uF
U4
STM32C011F4U6
VDD
VSS
PF2
PC14
PC15
PB6
PB7
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9/PA11
PA10/PA12
PA13
PA14
MCU_NRST
LED_K
SWDIO
SWCLK
C14
100nF
Bus map: ToF 0x29 (VL53L1X default), flow bridge 0x3B.
The bridge can hold the ToF in reset (XSHUT) to change its address.
D — Status LED, SWD pads, mounting
R11
1k
D1
green
TP1
TP2
TP3
TP4
TP5
H1
M2
H2
M2
Generated automatically: run ERC before release
PixArt PMW3901MB (flow, SPI) behind an ST STM32C011 I2C bridge; ST VL53L1X ToF on the bus
Qwiic (JST SH 4P) in/out: daisy-chain with the GPS board; I2C 0x29 (ToF) and 0x3B (flow bridge)
Mounted under the battery tray, sensor side facing the ground (handoff doc for the frame)
ACwing — Andrea Cirillo
Sheet: /
File: ACwing_Flow_A.kicad_sch
Title: ACwing Flow rev A — optical flow + ToF expansion for ACwing FC rev A
Size: A4
Date: 2026-11-02
Rev: A
KiCad E.D.A. 9.0.8
Id: 1/1